

Software Requirements Specification (SRS)

Shipment Tracking System

Document Name	Shipment Tracking System Specification
Linked PRD	Shipment Tracking System PRD
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System Overview

The system provides a comprehensive shipment tracking platform enabling real-time visibility into logistics operations across multiple carriers and modes of transport.

System Scope

This specification covers the backend services, API layer, real-time event processing pipeline, and integration interfaces required for the shipment tracking subsystem.

System Architecture Overview

The system follows a microservices architecture deployed on AWS ECS with an event-driven communication layer using Amazon EventBridge. Frontend communicates via REST and WebSocket APIs.

Functional Requirements

FR-001: Real-time shipment location updates with maximum latency of 30 seconds.

FR-002: Multi-carrier tracking through unified API interface.

FR-003: Automated alerts for shipment delays.

FR-004: Complete audit trail for shipment status changes.

Non Functional Requirements

NFR-001: Handle 10,000 concurrent tracking requests.

NFR-002: API response time under 200ms for 95th percentile.

NFR-003: 99.95% system uptime.

NFR-004: AES-256 encryption for data at rest.

System Interfaces

REST API for frontend, WebSocket for real-time updates, Webhooks for carrier events, Message queue for async processing.

API Requirements

Versioned REST endpoints, OAuth 2.0 authentication, JWT tokens, rate limiting 1000 requests per minute per client.

Data Requirements

Shipment retention 7 years, caching TTL 60 seconds, partitioning by tenant ID, PII tokenization in non-production.

Security Requirements

TLS 1.3 encryption, RBAC access control, input validation protection, tamper-proof audit logging.

Performance Requirements

Page load under 2 seconds, optimized database queries, background jobs processed within 5 seconds.

Scalability Requirements

Horizontal autoscaling, database read replicas, stateless services, event processing up to 50k events per second.

Operational Constraints

CI/CD deployment with rollback, Terraform infrastructure, centralized logging via ELK.

Assumptions

Carrier APIs maintain backward compatibility. AWS high availability. Modern browser support.

Dependencies

AWS ECS, RDS, Redis, PostgreSQL, carrier APIs, Auth0.

Risks

Carrier API rate limits, legacy data migration issues, multi-tenant isolation risks.